

BDX Slow Control - Alarm Reference

Global

Alarm	Condition	Level
Global interlock	INTERLOCK_ACTIVE = 1	INTERLOCK immediate
System not ready	READY = 0 and SYSTEM_STATE is neither STANDBY nor SAFE	MINOR 5 s; MAJOR 120 s
Global heartbeat stale	HEARTBEAT unchanged	MINOR 10 s; MAJOR 60 s

Chiller

Alarm	Condition	Level
Temperature deviation	RUN_RBV = On, not settling, temperature deviates from setpoint	MINOR $\geq 5\%$ for 5 s; MAJOR $>10\%$ immediate or $\geq 5\%$ for 20 s
Communication lost	COMM_OK = 0	MINOR 10 s; MAJOR 60 s
Heartbeat stale	HEARTBEAT unchanged	MINOR 10 s; MAJOR 60 s
IOC not running	IOC_STATE != RUNNING	MAJOR 5 s
IOC error	ERROR_CODE != 0	MINOR 10 s; MAJOR 60 s
Coolant level low	STAT_LOW_LEVEL = 1	MAJOR 60 s; INTERLOCK 600 s
Overtemperature	STAT_OVERTEMPERATURE = 1	MAJOR 60 s
General error/alarm/warning	Corresponding STAT bit = 1	MINOR 10 s
Reserved STAT bit	STAT_RESERVED = 1	MINOR 10 s
External control missing	STAT_EXTERNAL_CONTROL_MISSING = 1	MINOR 10 s

Low-voltage power supplies - LV1/LV2, CH1/CH2

Alarm	Condition	Level
Communication lost	COMM_OK = 0	MINOR 10 s; MAJOR 60 s
Heartbeat stale	HEARTBEAT unchanged	MINOR 10 s; MAJOR 60 s
IOC not running	IOC_STATE != RUNNING	MAJOR 5 s
IOC error	ERROR_CODE != 0	MINOR 10 s; MAJOR 60 s
Voltage deviation	OUTPUT On; $ V-RBV - V-set \geq 30$ mV and deviation $\geq 5\%$	MINOR 10 s; MAJOR 60 s
Output mismatch	OUTPUT_RBV != OUTPUT_SET after next hardware poll	MAJOR 5 s
OVP proximity	OUTPUT On; OVP ≥ 30 mV; V $\geq 95\%$ OVP / $\geq 100\%$ OVP	MINOR 5 s / MAJOR immediate
OCP trip	OCP_TRIPPED = 1	MINOR immediate; MAJOR 10 s; INTERLOCK 20 s
Current change 2 s	Current change $\geq 10\%$ and $ dl \geq 30$ mA within 2 s	MAJOR immediate
Current change 10 s	Current change $\geq 10\%$ and $ dl \geq 30$ mA within 10 s	MINOR immediate
Voltage span 30 s	VOLTAGE_SPAN_30S >0.1 V / >0.5 V	MINOR / MAJOR immediate

Temperature sensors - T00/T01/T02/T03

Alarm	Condition	Level
Outside monitoring range	Temperature <15 C or >28 C	MINOR immediate
Critical temperature	Temperature <10 C or >35 C	MAJOR 300 s
Fast temperature change	change >=3 C within 10 min	MAJOR immediate
Long temperature change	change >=5 C within 1 h	MAJOR immediate
Sensor unavailable	STATUS_OK = 0	MINOR 10 s; MAJOR 60 s
Heartbeat stale	HEARTBEAT unchanged	MINOR 10 s; MAJOR 60 s
Temperature spread	Max-min sensor spread >4 C	MINOR 60 s; MAJOR 1200 s

Humidity H01 / Pressure P01

Alarm	Condition	Level
Sensor unavailable	STATUS_OK = 0	MINOR 10 s; MAJOR 60 s
Heartbeat stale	HEARTBEAT unchanged	MINOR 10 s; MAJOR 60 s

Archiver

Alarm	Condition	Level
Archiver unhealthy	ARCHIVER:OK = 0	MINOR 10 s; MAJOR 60 s
Health check stale	LAST_CHECK unchanged	MINOR 30 s; MAJOR 60 s
Required PVs missing	REQUIRED_HEALTHY < REQUIRED_TOTAL	MINOR 10 s; MAJOR 60 s

Optional HV - HV1

Alarm	Condition	Level
Communication lost	COMM_OK = 0	MINOR 10 s; MAJOR 60 s
Heartbeat stale	HEARTBEAT unchanged	MINOR 10 s; MAJOR 60 s
IOC not running	IOC_STATE != RUNNING	MAJOR 5 s
IOC error	ERROR_CODE != 0	MINOR 10 s; MAJOR 60 s
Voltage deviation	OUTPUT On; V-RBV - V-set >=30 mV and deviation >=5%	MINOR 10 s; MAJOR 60 s
Output mismatch	OUTPUT_RBV != OUTPUT_SET after next hardware poll	MAJOR 5 s
OVP proximity	OUTPUT On; OVP >=30 mV; V >=95% OVP / >=100% OVP	MINOR 5 s / MAJOR immediate

Optional DAQ

Alarm	Condition	Level
Communication lost	COMM_OK = 0	MINOR 10 s; MAJOR 60 s
Heartbeat stale	HEARTBEAT unchanged	MINOR 10 s; MAJOR 60 s
IOC not running	IOC_STATE != RUNNING	MAJOR 5 s
DAQ error	ERROR field non-empty	MINOR 5 s; MAJOR 20 s
State mismatch	STATE_RBV != EXPECTED_STATE_RBV	MINOR 5 s; MAJOR 20 s
Not ready during run	READY = 0 while MEASUREMENT_ACTIVE = 1	MAJOR immediate